# File permissions in Linux

## Project description

In this project, I will examine permissions on the Linux file system to make sure the correct permissions are assigned to users, groups, and others. If I identify permissions that do not match the organization's authorization requirements, I will modify them to remove unauthorized access and ensure that only the appropriate users have access.

## Check file and directory details

I used the `ls -la` command in the `/home/researcher2/projects` directory to examine the permissions assigned to all files and directories, including hidden files. The `-l` option displays detailed information such as permissions, ownership, and group information, while the `-a` option includes hidden files. I reviewed the results to determine whether the existing permissions matched the organization's authorization requirements.

## Describe the permissions string

For example, `-rw-rw-rw-` represents the original permissions for `project_k.txt`. The first character `-` indicates that it is a regular file. The next three characters `rw-` show that the user has read and write permissions. The following `rw-` shows that the group also has read and write permissions. The final `rw-` shows that others also have read and write permissions. A dash (`-`) means that a specific permission has not been granted.

## Change file permissions

I used the `chmod` command to change the permissions on `project_k.txt`. In the command `chmod o-w project_k.txt`, `o` represents others, the minus sign (`-`) means that a permission is being removed, and `w` represents write permission. This removed write access from others because the organization does not allow others to have write permission on files. After making the change, I used `ls -la` to verify that the permission had been removed.

## Change file permissions on a hidden file

I used `chmod u-w,g-w,g+r .project_x.txt` to change the permissions on the hidden file `.project_x.txt`. The `u-w` removed write permission from the user, `g-w` removed write permission from the group, and `g+r` added read permission to the group. These changes ensured that the user and group could read the archived file but no one had write permission. I used `ls -la` afterward to verify the new permissions.

## Change directory permissions

I used `chmod g-x drafts` to remove the group's execute permission from the `drafts` directory. The `g` represents the group, the minus sign (`-`) removes a permission, and `x` represents execute permission. Removing this permission prevents the group from accessing the directory, leaving only the owner, `researcher2`, with access. I used `ls -la` afterward to verify the permission change.

## Summary

I reviewed the file and directory permissions and made changes based on the organization's authorization requirements. For `project_k.txt`, I removed write permission from others using `chmod o-w project_k.txt`. For the hidden file `.project_x.txt`, I removed write permission from the user and group and added read permission for the group using `chmod u-w,g-w,g+r .project_x.txt`. Finally, I used `chmod g-x drafts` to remove the group's execute permission so that only `researcher2` could access the `drafts` directory. I verified the permission changes using `ls -la`.